

CXR Langfuse Lab Manual
SW.18 — LLM traces + full wiring guide

CXR Ops Lab — `cxr-ops-lab`

2026-05-31

Abstract

SW.18: Langfuse v2 + Postgres on **:3100**. This manual documents **exactly how it is wired** — Compose → UI signup → API keys → `send-trace.mjs` → Traces tab — and where it would attach to Claim Studio judge / doc-reasoning routes in a future phase (not live today).

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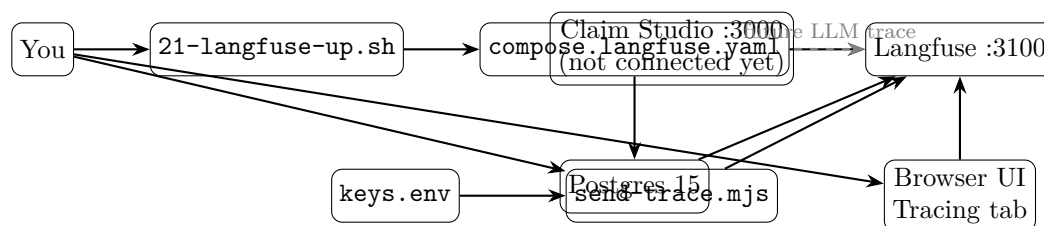
Chapter 1

How this lab is wired (step by step)

1.1 The chain (what runs when you type the commands)

1. You run `./scripts/21-langfuse-up.sh`
2. Script calls `docker compose -f compose.langfuse.yaml up -d`
3. `compose.langfuse.yaml` starts two containers:
 - `langfuse-postgres` — Postgres 15, volume `langfuse_pg_data`
 - `langfuse` — image `langfuse/langfuse:2`, host port **3100** maps to container **3000**
4. First browser visit `http://localhost:3100` — sign up (local OSS instance)
5. UI: + New Organization → create project (e.g. `cxr_claims`)
6. Settings → API Keys — create keys → paste into `lab/langfuse/keys.env`
7. You run `node lab/langfuse/send-trace.mjs` (or `./scripts/21-langfuse-smoke.sh`)
8. Script loads `keys.env`, uses Langfuse Node SDK, POSTs trace `cxr-sw18-golden-path`
9. UI: Tracing → Traces — golden path appears with mock generation

1.2 Wiring diagram



Chapter 2

What is wired vs not wired

Layer	Status today	Future (sketch)
Langfuse + Postgres :3100	Running via Compose	Managed Langfuse Cloud or v3 stack
Project API keys	User-created in UI	CI secret / K8 Secret
Golden trace ingest	<code>send-trace.mjs</code>	Real judge / analyzer spans
Claim Studio :3000	Not connected	<code>analyze/route.ts</code> emits Langfuse spans
Rehearsal :8251	Not connected	OTel + Langfuse dual export
Jaeger SW.11	Separate	Traces vs LLM evals (complementary)

Today this lab proves *local LLM observability* and the *golden trace name* `cxr-sw18-golden-path` for syllabus M7.8. Claim Studio still runs without Langfuse SDK.

Chapter 3

Why v2 not v3

Bootcamp uses **Langfuse v2 + Postgres only** — lighter than v3 (ClickHouse, MinIO, Redis, worker). Port **3100** avoids collision with CXR Claim Studio on **:3000**.

Chapter 4

Golden trace (what `send-trace.mjs` sends)

Field	Value
Trace name	<code>cxr-sw18-golden-path</code>
Input	<code>{ feature: doc-reasoning-sketch, claim_id: demo-1 }</code>
Metadata	<code>{ lab: SW.18, syllabus: M7.8 }</code>
Generation	<code>mock-llm-call / model bootcamp-mock</code>
Host default	<code>http://127.0.0.1:3100 (override via LANGFUSE_HOST)</code>

Chapter 5

UI navigation (verified bootcamp path)

1. Open `http://localhost:3100` — Home **Get Started**
2. + **New Organization** (e.g. org `cxr`)
3. Create project (e.g. `cxr_claims`)
4. **Settings** → **General** — confirm host `http://localhost:3100`
5. **Settings** → **API Keys** — create public + secret key
6. Copy keys to `lab/langfuse/keys.env` (**not** `keys.env.example`)
7. Run trace script; then **Tracing** → **Traces** → `cxr-sw18-golden-path`

Chapter 6

keys.env format (common pitfall)

The trace script reads **only** `lab/langfuse/keys.env`. Lines must be **uncommented**:

```
LANGFUSE_HOST=http://127.0.0.1:3100
LANGFUSE_PUBLIC_KEY=pk-lf-...
LANGFUSE_SECRET_KEY=sk-lf-...
```

`keys.env` is gitignored. If `send-trace.mjs` prints `SKIP`, the file is missing or keys are still commented in `.example`.

Chapter 7

Files in the repo

File	Role in the wiring
<code>compose.langfuse.yaml</code>	Postgres + Langfuse v2, port 3100
<code>scripts/21-langfuse-up.sh</code>	Wrapper: <code>compose up -d</code>
<code>scripts/21-langfuse-smoke.sh</code>	HTTP :3100 + optional trace ingest
<code>lab/langfuse/send-trace.mjs</code>	Langfuse SDK golden trace
<code>lab/langfuse/keys.env.example</code>	Template (copy → <code>keys.env</code>)
<code>lab/langfuse/package.json</code>	langfuse npm dependency
<code>evidence/SW18-langfuse-verify-2025-05-31.md</code>	Ch2025-05-31

Chapter 8

Hypothetical Claim Studio wiring (not deployed)

```
# Future sketch only:
# 1. npm install langfuse in cxr-ui rehearsal app
# 2. In judge/analyze route: Langfuse client with env from keys
# 3. trace({ name: "cxr-judge", input: claimId, ... })
# 4. generation({ model, prompt, output }) for each LLM call
# 5. Same project keys as bootcamp lab for local dev
```

Chapter 9

Live adjustments

- **Port 3100** — deliberate; do not point Langfuse at :3000.
- **v2 image pin** — `langfuse/langfuse:2`; v3 needs extra services.
- **First signup** — data persists in Postgres volume until `compose down -v`.
- **keys.env vs .example** — script ignores commented template file.
- **Smoke without keys** — UI health still passes; trace step SKIPS gracefully.
- **localhost vs 127.0.0.1** — both work if `NEXTAUTH_URL` matches browser URL.

Chapter 10

Procedure

```
cd cxr-ops-lab
./scripts/21-langfuse-up.sh
# Browser: http://localhost:3100
# Sign up -> org -> project -> Settings -> API Keys
cp lab/langfuse/keys.env.example lab/langfuse/keys.env
# Edit keys.env: uncomment and paste pk/sk
node lab/langfuse/send-trace.mjs
./scripts/21-langfuse-smoke.sh
# Tracing -> Traces -> cxr-sw18-golden-path
```

Chapter 11

Stop

```
docker compose -f compose.langfuse.yaml down  
# Add -v to wipe Postgres (signup + traces)
```

Chapter 12

Bootcamp context

SW.12–SW.18 electives live in `cxr-ops-lab` as **standalone Compose stacks**. Master port map and cross-lab wiring: `docs/CXR-BOOTCAMP-LABS-COMPENDIUM.pdf`.